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Characterization of Surface Roughness Effects on Pressure Drop in Single-phase Flow in Mini/Micro Channels

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ABSTRACT

The roughness features on the wall affect the pressure drop in the fluid. Their effect on the flow is manifested by (i) flow area constriction, and (ii) increase in the wall shear stress due to flow over the roughness features. Applying the area constriction to the Moody's chart results in a simplified plot that yields an asymptotic value of friction factor for higher relative roughness values in the turbulent region. Specific experiments were conducted to study the effect of surface roughness characteristics on pressure drop using parallel saw-tooth elements in aligned and offset configurations in rectangular channels with different channel heights. The experimental results indicate that the use of constricted flow cross-sectional area, based on $D_{cf} = D_t - 2\varepsilon$, in the modified Moody plot is able to extend the applicability of the laminar flow friction factor equation to the relative roughness values (defined as the saw-tooth height) of up to 14%. In the turbulent region, there is a need to further characterize the surface features beyond the average roughness,

R_a , which is used extensively in the fluid mechanics and convective heat transfer applications. A number of available roughness characteristics are reviewed for surface characterization in minichannels and microchannels, where large relative roughness values may be encountered.

INTRODUCTION

The effects of surface roughness on pressure drop in circular and rectangular tubes have been studied extensively in literature. As early as in the nineteenth century, Darcy [1] conducted careful pressure drop experiments on pipes of different materials and roughness, and introduced the concept of relative roughness. He also established that the flow depended on the pipe roughness, pipe diameter, and slope. Fanning [2] later proposed a correlation for the pressure drop as a function of surface roughness. The pioneering work of quantifying the effect of surface roughness on pressure drop was led by Nikuradse [3], who established the sand grain roughness, equivalent to the diameter of the sand particles, as a major parameter in defining the friction factor during laminar and turbulent flow studies. To create the roughness on the pipes, he first sifted sand grains so that the grains were of uniform size. He then used Japanese lacquer to stick the sand grains to the walls of the pipe. Through exhaustive experimentation, Nikuradse studied the effect of roughness on flow characteristics in circular pipes. Experimental data was obtained for water flowing in pipes of six different relative roughness surfaces with Re ranging from 600 to 10^6 . The dimensions of the inner diameters of the test pipes were 25, 50, and 100 mm. His work helped establish the effect of relative roughness (ε/D_t), on the flow characteristics. It should be noted that the pipe diameter used in his work was the base diameter of the pipe before applying the sand grains.

Colebrook [4] later published his findings on the variation in surface roughness and its effects on pressure drop. Colebrook's experiments included sixteen spun concrete-lined pipes and six spun bitumastic-lined pipes ranging in diameter from 4 inches to 60 inches. Average relative roughness values ranged from 0.04318 mm to 0.254 mm. Also

tested was the 216-inch diameter Ontario tunnel. The Ontario tunnel was a concrete tunnel made from steel oil forms. When the concrete set and the forms were removed, the walls of the tunnel were rubbed down with carborundum brick to remove any deformities. These pipes were tested in the transition and fully turbulent regions. It was determined that the friction factor decreases as the velocity increases. Once the velocity profile reaches the fully developed profile described by the square law, the friction factor remains constant for all higher velocities.

Moody [5] later characterized the friction factor (f) as a function of the Reynolds number and relative roughness (ε / D_t). Through Colebrook's work, the Moody diagram was established which provided engineers with a convenient method of determining the pressure drop associated with the relative roughness of the pipe. The outlined method of determining pressure drop loses its integrity when the surface roughness is on the same order of magnitude as the characteristic diameter of the pipe. Although the current relative roughness values used in the plots go only as high as 5 percent, higher values are expected while dealing with flows in small diameter channels. The sand grain diameters in Nikuradse's experiments were used for roughness ε . The average roughness value, R_a , obtained from surface profile measurements were later employed by researchers and designers while using the Moody diagram.

Webb, et al. [6] studied the separation of flow from the wall of a pipe when passed over repeating roughness elements. Flow was determined to separate from the wall following roughness elements and reattach at a distance of 6-8 times the height of the element. When the roughness elements are the result of a manufacturing process such as a cutting blade or surface treatment, the elements are similar in nature and closely spaced.

The flow encounters the leading element and skims over a pocket of circulating liquid before passing over the following element.

In the present paper, the constriction effect on the Moody diagram is plotted by considering the constricted flow area in the turbulent region. Further, to ascertain the effect of surface roughness features on the friction factor, systematic experiments are conducted using air and water as working fluids. The need for a new set of surface roughness descriptors for fluid flow characterization is then discussed.

FLOW CONSTRICTION EFFECT ON FRICTION FACTOR

The Moody diagram, shown in Figure 1, is simply the Colebrook-White equation plotted over a wide range of Reynolds numbers and relative roughness values. It serves as an easy way for engineers to estimate the friction factor without having to perform a calculation for the friction factor. By calculating the Reynolds numbers and relative roughness value, an engineer can quickly find a pipes friction factor. By finding where the calculated Reynolds number intersects the relative roughness line for a particular roughness and tracing the intersection back to the friction factor on the y axis, the friction factor of a given flow may be determined with minimal effort.

As seen from Fig. 1, the laminar friction factor is independent of the relative roughness in the range $\varepsilon/D_t \leq 0.05$. In the turbulent region, the friction factor for smooth tubes continuously decreases with increasing Reynolds numbers. For rough tubes, the friction factor follows the smooth tube plot up to a certain point, and then approaches an asymptotic value depending on the relative roughness. The asymptotic value increases with relative roughness as shown in Fig. 1 in the range $0 \leq \varepsilon/D_t \leq 0.05$.

The above trend in friction factor in the turbulent region can be explained as follows: For a given relative roughness, the roughness features do not affect the flow up to a certain value of Reynolds number. For higher Reynolds number values, the surface effects are felt in the flow, eventually reaching an asymptotic friction factor value.

The calculations of Moody assumed that the flow boundary occurred at the wall of the tube and utilized the full diameter for flow area. Following the discussion presented by Webb et al. [6], Fig. 2 is drawn to show the effects of roughness elements on the reattachment of flow to the pipe walls for different roughness heights. It can be seen from Fig. 2 that for low roughness values, the flow reattaches to the tube wall. At higher roughness values, the flow does not re-attach to the pipe wall, thus decreasing the available pipe area for fluid flow. The new flow boundary is suspended above the surface wall at a distance y . The diameter of the flow, then, is not the diameter of the pipe, but is reduced by the size of the roughness elements. While the exact value of this height in comparison to the height of the roughness elements is not known, it is bounded by $y=0$ and the height of the elements. In closely spaced roughnesses, such as the sand grains used in Nikuradse's experiments, the value of y closely resembles ε . The new flow diameter can then be approximated by:

$$D_{cf} = D_t - 2\varepsilon \quad (1)$$

where D_{cf} is the constricted flow diameter, D_t is the tube (root) diameter, and ε is the average roughness height. Drawing similarity to the compact heat exchanger geometries, it may be noted that the resulting area based on D_{cf} corresponds to the free flow area.

The decrease in flow diameter can have a strong influence on the friction factor used. Expressing friction factor in terms of mass flow, pipe diameter and pressure gradient for

fully developed flow in a circular pipe, the friction factor used in the Moody plot is given by:

$$f_{Moody} = \left(\frac{\frac{dP}{dX} \rho \pi^2}{8 \dot{m}^2} \right) D_t^5 \quad (2)$$

The Moody diagram is plotted using the following equation by Colebrook [4].

$$\frac{1}{f_{Moody}^{0.5}} = -2.0 \log \left(\frac{\frac{\varepsilon}{D_t}}{3.7} + \frac{2.51}{\text{Re } f_{Moody}^{0.5}} \right) \quad (3)$$

The relation between f_{Moody} and f_{cf} is given by:

$$f_{cf} = f_{moody} \left[\frac{(D_t - 2\varepsilon)}{D_t} \right]^5 \quad (4)$$

The Reynolds number is also modified using the constricted flow diameter and is given by:

$$\text{Re}_{cf} = \frac{4 \dot{m} \pi D_{cf}}{\mu} \quad (5)$$

Substituting eqs. (1) and (4) in eq. (3), we get,

$$\frac{1}{\left(\frac{D_t}{D_t - 2\varepsilon} \right)^{2.5} f_{cf}^{0.5}} = -2.0 \log \left(\frac{\frac{\varepsilon}{D_t}}{3.7} + \frac{2.51}{\text{Re}_{cf} \left(\frac{D_t}{D_t - 2\varepsilon} \right)^{3.5} f_{cf}^{0.5}} \right) \quad (6)$$

The modified equation for pressure gradient for a circular pipe is thus given by:

$$\frac{dP}{dx} = \frac{8\dot{m}^2 f_{cf}}{\pi^2 \rho D_{cf}^5} \quad (7)$$

Figure 2 shows the Modified Moody diagram plotted using eqs. (5) and (6) in the turbulent region with friction factor f_{cf} and Reynolds number Re_{cf} based on D_{cf} . It can be seen from Fig. 2 that the essential features of the Moody diagram are retained, in terms of the transition region leading to an asymptotic value of f at higher Reynolds number. However, a major change is seen in that the asymptotic value of f_{cf} , obtained for $\varepsilon/D_t > 0.03$ at higher Re_{cf} values, plateaus to approximately $f_{cf}=0.042$. In other words, for relative roughness values higher than 0.03, the friction factor f_{cf} based on the constricted diameter D_{cf} is constant in the fully rough region. This means that the roughness effects on the friction factor are completely accounted for through the area constriction effects incorporated in f_{cf} . Intuitively, this makes sense as a higher roughness feature would result in a higher flow constriction. This corresponds to the region described by Webb et al. [6] where the flow does not reattach to the walls as shown in Fig. 2.

It is strongly advised that the above plots are valid only in the range $0 \leq \varepsilon/D_t \leq 0.05$ as in the original Moody diagram. There is a need for reliable experimental data in the region $\varepsilon/D_t > 0.05$ for circular pipes before the Moody diagram can be extended to pipes with higher relative roughnesses. The effect of the constriction in the laminar region is not as pronounced, especially for $\varepsilon/D_t \leq 0.05$. For higher values of relative roughness, experimental confirmation is needed in both laminar and turbulent ranges.

In using the Moody diagram, one would measure the tube inner diameter, which in reality represents D_{cf} and use it as D_t in pressure drop calculations. In Nikuradse's experiments, the base diameter D_t was employed in the pressure drop calculations.

Although this error is quite small for the range of roughness employed in the Moody diagram, it could become significant if the diameter correction is not incorporated while extending the Moody diagram for pipes with higher roughness values. The fact that f_{cf} remains constant at 0.042 in the high Reynolds number region also simplifies the calculations to some extent. It should be noted that in calculating the pressure drop, the actual constricted flow area should be used as given by eq. (5).

ROUGHNESS EFFECT ON FRICTION FACTOR

The roughness imparted on the internal walls of a flow channel are dependent on the manufacturing processes. The sand grain roughness feature is used extensively in fluid mechanics application. With smaller channel sizes, the relative roughness could be significantly greater than the threshold of 0.05 used in the Moody diagram. A survey of the manufacturing processes is presented first, followed by the relevant roughness parameters that hold promise. Finally, the effect of high relative roughness and one specific roughness feature on the friction factor in the laminar and transition region is investigated.

SURVEY OF MANUFACTURING PROCESSES

Process develops surface geometry and roughness

All manufactured products are created with specific materials and geometry to accomplish the function for which they were intended. The specific geometry of parts can be created via any of the major subsets of processes; phase change processes,

deformation processes, material addition processes, or material removal processes. All processes have a direct effect on material properties and vice versa. The most typical scenario in manufacturing is a type of balancing act in selecting materials and processes that come together with a proper design to generate the desired resulting product.

Phase-change processes

Casting and molding processes generate part geometry via a phase-change from liquid to solid. The process is extremely efficient and is typically adapted to high volume production processes. The most typical drawback to casting and molding processes is a lack of dimensional control and poor material properties. The thermal contraction of materials and the volume contraction associated with the liquid to solid phase change causes severe difficulties when working at a mini or micro scale. The difficulty of holding any dimensional tolerances with complex geometries is usually prohibitive at a very small scale. If dimensional properties can be maintained, the poor material strength associated with casting and molding due to the properties and the direction of grain growth is typically prohibitive.

At the small scale there are also some deposition and plating processes that make use of vapor or liquid to solid transformations, typically for coating applications. These processes are typically included in material addition processes and will be discussed further in that section below. Complex geometry and minute features are not possible for mechanical structures in vapor deposition processes, but masking and etching

technologies do make such processes candidates for simple structures and geometry generation.

Deformation processes

Macro scale processes such as forging, punching, and rolling are typically not utilized at the mini or micro scales, though some examples do exist. This class of processes makes use of plastic deformation to change the size and shape of materials until they meet design requirements. The materials experience severe strains and typically undergo strain hardening that can drastically change material properties. At the small scale these are often prohibitive characteristics, but some examples of successful processes do exist. Micro wire drawing processes are routinely used to make very fine metal wire for external connectivity and other applications.

Material addition processes

Electronics processing and rapid prototyping technologies have made use of manufacturing processes that actually add material to the part rather than removing it to generate the desired geometry. This style of process has the advantage that material is only deposited where it is needed and is therefore very efficient from a time and energy usage perspective. The main drawback of such processes is the poor material properties of the material that is deposited due to interfacial bonding issues. Control of bonding to the initial substrate and crystal structure growth are typically very poor in material addition processes and the result is very poor mechanical performance of the parts and somewhat compromised electrical properties. Such processes are often necessary

because there is no other way to accomplish the required design, but are typically avoided in most situations.

Material removal processes

Machining is the most developed of all processes and very controlled dimensions and material properties can be achieved with most materials in use today. Many different methods for material removal have been developed including rotating, abrasive, laser, water jet, acid etch, electrical discharge, and other removal processes. All material removal processes work as a type of sculpting process, methodically removing material until the desired part geometry is obtained from a larger block of material. All material removal processes generate the gross geometry of the part with relative success, but the fine features of the surface that is generated vary a great deal depending on the material removal method used. As the feature sizes of the parts are reduced the relative importance of the surface features is amplified, creating not only a texture, but actual features of a size that is relevant on the part surface.

Traditional machining

Traditional material removal processes make use of the rotation of a spindle to generate relative motion between the work part and the tool. In the case of a cylindrical component, the part is rotated in the spindle and the tool is pseudo stationary as it generates the desired part profile. In the case of prismatic components, the part is held stationary while the tool is rotated in the spindle and forced to interfere with the part in such a way as to generate the desired part geometry. The nature of the material removal

process is such that the tools will remove material by following paths that overlap to some degree (but not completely) to generate the gross geometry of the part and as a byproduct the fine features of the surface. For the purposes of this paper, it is assumed that the gross geometry of the parts is attainable without difficulty, but it is the relative size of the fine surface features (tool marks) with respect to the devices being fabricated that are of most concern because they will in fact have drastic impacts upon the performance of the devices. With traditional spindle-based machining processes fine tool marks occur at regular intervals (typically related to the feed rate of the tool) and have a geometry that is the inverse of the tool tip. A typical surface generated in a turning process is shown in Figure 3. In the mini / micro range the relative size and impact of these surface features is increased dramatically.

Nontraditional material removal processes

The past few decades have seen the development of a host of material removal processes that do not make use of a rotating spindle and tool in the traditional manner. Processes that fall under this category include laser machining, water jet cutting, plasma arc cutting, ultrasonic machining, etc. One of the most common of these processes is electrical discharge machining (EDM) in which a surface is eroded via electrical arcing from a tool to the part. As the process progresses the gross geometry of the part takes on the inverse shape of the tool that is used to generate it. The process is commonly used for complex die cavity generation because the process can use relatively soft and easily machinable graphite or copper electrodes to machine extremely tough tool steel parts with very complex geometry. The fine surface features of this process, as well as most

nontraditional processes are more random in frequency, size, and shape. Figure 4 shows the surface features of a part that was created using EDM.

Etching processes

Another set of material removal processes relies on chemical means to remove material and have become of critical importance in electronic manufacturing. Acid/chemical etching and electrically assisted etching techniques are effective in removing material, but great care must be taken to generate the desired geometry by masking or other means. Etching processes are particularly sensitive to crystal structures and grain boundaries that cause variations in the etching rates at different places on the same part. In general etched surfaces take on the features and character of the underlying material structure. These features are typically magnitude limited only by the time and extent of the etching process.

Micro fabrication processes

The advent of electronics and their ever-decreasing feature sizes has had a strong impact on the development of processes to support such fabrication. This set of processes has become known as micro fabrication technology, but is actually made up of a series of different processes. Layered integrated circuits have caused the development of both material removal and material addition processes in addition to doping processes which change the electrical properties of the parts. The gross and fine geometry produced by the removal and addition processes is typically layered in nature, but does consist of full 3-D features. Figure 5 shows the gross geometry of a silicon substrate after processing,

and the chips on the surface. The fine features of the micro-fabricated parts are generated by a series of masking, etching, and deposition processes that perform according to the descriptions above. The general character of the features can often be exaggerated due to the large number of sequential steps involved in the production of layered IC chips and other micro fabricated parts. Figure 6 shows the typical dimensional range of micro fabrication processes as compared to conventional machining processes.

Surfaces

The assessment of surfaces is a difficult and confusing proposition for most situations in manufacturing and production environments. The link between design intent and the performance of a given surface is usually tenuous at best and is most often based on experience and feedback from field experience, which is clearly post-manufacturing. In general manufacturing and production strive to provide what the designer has called out whether the designer had a good reason to call it out or not. A big part of this process is the measurement and characterization of surfaces. Most typically a surface is characterized by gross, medium, and fine features separately. Gross features are typically characterized by dimensions and tolerances. Medium features typically are characterized as large wavelength surface variations known as waviness or flatness. Fine surface features are typically described as high frequency variations in the surface, better known as roughness. Figure 7 shows the difference between medium and fine surface features. Characterization of surfaces is typically done on a worst case scenario basis, measuring variations across the lay of the part, the lay typically being tool marks.

EFFECT OF ROUGHNESS FEATURES ON FRICTION FACTOR

The Moody diagram is applicable to pipes that have relatively low values of relative surface roughness. The underlying data in this diagram is based on sand grain roughness. This diagram is used extensively in predicting friction factors for commercial tubes whose roughness features are not too different from the sand grain roughness. As the relative roughness becomes large, it becomes important to consider the roughness features as well.

Need for roughness characterization

The pioneering work on characterizing the effect of surface roughness on pressure drop by Nikuradse [1] was achieved by using sand to induce known roughnesses on flow pipes. This work provided the foundations for establishing the effects of relative roughness (e/D) on flow characteristics. However, this characterization of surface roughness was limited to the diameter of the sand grain (equivalent to the modern descriptor *Average Maximum Height of the Profile, R_z*) and did not consider other parameters that would describe the spacing, waviness, and shape in a three-dimensional area profiling. Additionally, using the definitions by Kandlikar and Grande [7], when considering minichannels ($3 \text{ mm} \geq D_h \geq 200 \text{ }\mu\text{m}$) and microchannels ($200 \text{ }\mu\text{m} \geq D_h \geq 10 \text{ }\mu\text{m}$), the ratio between the hydraulic diameter and the geometry of the roughness profile becomes significantly different from those in conventional channels. At this scale, the shape, spacing and size of the roughness irregularities have different influence upon the pressure drop and the overall fluid regime. In particular, a more complete characterization may prove useful in attempting to describe the flow separation behavior exhibited in flow

patterns when encountering features from the surface topography. Hence, there is a need for developing a set of roughness descriptors suitable for micro and mini channels that would allow for a complete understanding the effect of different aspects of surface topography upon pressure drop and fluid behavior.

Surface characterization parameters and possibilities

In any given channel cross section there are two surfaces with roughness features of interest (see Figure 8). The proposed approach formalizes a set of descriptors to be collected from each of the two surfaces, as well as statistics on the hydraulic diameter from the cross section of interest describing the interaction between the two surfaces.

i) Characterizing the individual surfaces

A set of six parameters is proposed to provide information on the roughness features: average height (Rz), spacing (RSm and Pc), shape (Rsk and tp), as well as average deviations from mean line (Ra). These parameters are defined in the ASME B46.1-2002 [8] as follows

- Average Roughness (Ra): the arithmetic average of the absolute values of the profile height deviations (Z_i). See Figure 9.

$$Ra = \frac{1}{L} \int_0^L |Z(x)| dx \quad (8)$$

- Mean Spacing of Profile Irregularities (RSm): consists of the mean value of the spacing between profile irregularities within the evaluation length. See Figure 10.

$$RSm = \frac{1}{n} \sum_{i=1}^n Sm_i \quad (9)$$

- Peak Density (Pc): the number of SAE peaks per unit length measured at a specific peak count level over the evaluation length. See Figure 11.
- Skewness of Profile (Rsk): a measure of the asymmetry of the profile about the mean line calculated over the evaluation length. See Figure 12.

$$Rsk = \frac{1}{Rq^3} \frac{1}{L} \int_0^L Z^3(x) dx \quad (10)$$

where Rq is the root mean square average of the profile deviations from the mean line.

- Profile Bearing Length Ratio (tp): the ratio of the profile bearing length to the evaluation length at a specified level p. See Figure 13.

$$tp = \frac{b_1 + b_2 + b_3 + \dots + b_n}{L} \times 100\% \quad (11)$$

- Average Maximum Height of the Profile (Rz): the average of successive values of Rt_i calculated over the evaluation length, where $Rt_i = Rp_i + Rv_i$. See Figure 14.

ii) Characterizing the interaction between the surfaces

Additionally, after the two surfaces from a cross-section of a channel have been individually characterized, the combined effect of the surface features in the effective hydraulic diameter must be captured. The proposed approach suggests collecting the individual profiles and, via digital sampling, calculating the instantaneous hydraulic diameter along the evaluation length of interest for the channel. See Figure 15. Afterwards, three statistics are to be collected: Maximum Hydraulic Diameter (Max $|Dh_i|$), Minimum Hydraulic Diameter (Min $|Dh_i|$), and the Average Hydraulic Diameter measured from the filtered mean lines for each surface.

$$\overline{D}_h = \frac{1}{n} \sum_{i=1}^n |D_{h,i}| \quad (12)$$

In summary, the aforementioned approach incorporates measurement and assessment of six surface descriptors for each of the two individual surfaces that represent a two-dimensional cross-section, as well as three statistics that represent the impact of the roughness features in the interacting surfaces upon the localized reduction or expansion of the effective hydraulic diameter.

EXPERIMENTAL VERIFICATION OF THE ROUGHNESS AND FEATURE EFFECTS

The effect of roughness beyond $\varepsilon / D_t > 0.05$ is not studied in literature. The higher roughness is usually incorporated by incorporating surface features. The effect of feature size, shape and distribution on friction factor then becomes important. To verify this effect, Schmitt [9] performed an experimental study to evaluate the friction factor for a grooved surface in a rectangular minichannel. Two cases were investigated with the grooves in the top and bottom channel surfaces either aligned or offset in relation to each other. The details of the experimental setup and the test sections are given by Schmitt [9], and some essential features are described below.

Both air and water have been used as test fluids in the present work. Figures 16 and 17 show the respective experimental loops.

For the experiments with air, *zero air* is supplied in pressurized cylinders. The air is then passed through a bank of flow meters before entering the test section as shown in Fig. 16. The test section is 80 mm long and has thirteen pressure taps located along its length. For experiments with water, deionized (DI), degassed water is used as the test

fluid. The water is degassed using the procedure described in Kandlikar et al [10]. The DI, degassed water is drained into a Fisher Scientific Isotemp 3013 water bath. A positive displacement 1/4 HP bronze gear pump capable of pumping up to 0.9 GPM feeds the test section. A by-pass loop is used to control the flow rate through the test section. The water is then filtered using a Shelco FOS housing with a 1 μ m wound fiber filter. Three Omega FL-5500-NV rotameter flow meters control the flow rate to the test section.

The pressure taps from the test section are connected to a manifold through valves to facilitate differential pressure measurement, measured from the manifold inlet. Four pressure transducers with ± 1 psi, ± 5 psi, ± 15 psi, or ± 30 psi differential pressure ranges respectively are used to cover the entire operating range with air and water. K-type thermocouples measure the fluid temperature before and after the test section. Once the water passes through the test section, it is recirculated into the constant temperature water bath, while air is discharged into the atmosphere.

TEST SECTION DETAILS

Test section is made of two individual inserts, each 10 mm wide and 100 mm long, forming the top and bottom sides of the flow channel. They are mounted in a fixture that allows the gap between them to be controlled using two micrometers. A photograph of the assembly is shown in Fig. 18. The fluid inlet is through smooth entrances machined on the inserts. The surfaces of the inserts are machined to provide smooth or rough surfaces.

Three surfaces are tested in the current study. These surfaces are (i) smooth, (ii) aligned saw-tooth roughness, and (iii) offset saw-tooth roughness. The smooth channel is

shown in Figure 19. The channel is relatively smooth with a maximum average roughness height, R_z of $1.22\mu\text{m}$. The smooth minichannel is used to validate the test section since a smooth channel will not deviate from the conventional flow theory.

The aligned saw-tooth roughness arrangement is shown in Figure 20. It consists of a rectangular minichannel having parallel surfaces with machined ribs running perpendicular to the flow direction. The surface roughness peaks on one side of the channel are lined up to match the surface roughness peaks on the opposite side of the channel.

The offset saw-tooth roughness arrangement also consists of parallel surfaces with machined ribs running perpendicular to the flow direction, but the surface roughness peaks are lined up such that the point of one surface roughness peak on one side lies centered between the two surface roughness peaks directly across from it. This is shown in Figure 21.

The surface roughness parameters were measured using a Mitotoyo SJ-400 profilometer. The results are shown below.

ERROR ANALYSIS

The experimental uncertainty is given as:

$$U = \sqrt{B^2 + P_e^2} \quad (13)$$

where U is the uncertainty, B is the bias error, and P_e is the precision error. Systematic, or bias error is generally induced by the error in the measuring instrumentation. Bias can

be determined from the least count of the measuring instrumentation. Bias error is calculated by dividing the accuracy by the target value:

$$B = \frac{\text{Instrumentation Accuracy}}{\text{Measured Value}} \quad (14)$$

Precision error is error induced by a number of measurements. Precision is calculated as:

$$P_e = \pm t \times \frac{\sigma_s}{\sqrt{n}} \quad (15)$$

where t is determined from standard T-tables, σ_s is the sample standard deviation, and n is the number of samples. Temperature readings are accurate to $\pm 0.1^\circ\text{C}$, air and water flow rates are accurate to $\pm 3\%$ full scale, minichannel dimensions are accurate to $\pm 5 \mu\text{m}$, pressure readings are accurate to $\pm 0.25\%$ full scale, pressure tap distances are accurate to $\pm 5 \mu\text{m}$, and pressure tap locations are accurate to $\pm 5 \mu\text{m}$. The friction factors were found to have an uncertainty of 8.81%.

RESULTS

Tests were conducted for both air and water in minichannels with hydraulic diameters ranging from $325\mu\text{m}$ to $1819\mu\text{m}$ over a range of Reynolds numbers ranging from 200 to 7200 for air and 200 to 5700 for water. Static pressure was measured along the length of the minichannel in 13 locations placed uniformly along the length of the channel. The results for the different test fluids are discussed in the following sections.

Test Section Validation

The test section was fitted with smooth inserts and was run with air and water. The experimental results were compared with the theoretical results obtained using the following equations:

Laminar flow:

$$f = \frac{C}{\text{Re}} \quad (16)$$

Where the constant C depends on the aspect ratio $\alpha^* = b/w$. It is given by the following equation, Kakac et al. [11]:

$$C = 24 \left(1 - 1.3553\alpha^* + 1.9467\alpha^{*2} - 1.7012\alpha^{*3} + 0.9564\alpha^{*4} - 0.2537\alpha^{*5} \right) \quad (17)$$

In the turbulent region, the friction factor is given by the following equation, Miller [12]:

$$f = \frac{0.25 \left[\log \left(\frac{e/D_h}{3.7} + \frac{5.74}{\text{Re}^{0.9}} \right)^{-2} \right]}{4} \quad (18)$$

The applicability of the above equation with hydraulic diameter for rectangular channels was confirmed by Kakac et al. [11].

Figures 22 and 23 show a comparison of the experimental results obtained for the smooth channels with the theoretical results obtained from eqs. (16) - (18) for air and water respectively. The solid line represents the laminar flow and the dotted line represents the turbulent flow. It can be seen that the agreement is quite good, with a mean error of less than 3 percent. Transition from laminar to turbulent flow was observed around $\text{Re}=1950$. Similar agreement was noted with the water data.

EFFECT OF ROUGHNESS STRUCTURE

The experiments were conducted with two configurations with grooved inserts as shown in Figs. 20 and 21. The gap between the inserts was systematically varied to yield different ε/D_{cf} ratios. The hydraulic diameter was calculated using the constricted flow area, with width as w and the constricted gap $b_{cf} = b - 2\varepsilon$ (see Figs. 20 and 21).

Figures 24 - 27 show the friction factor variation with Re_{cf} for the two configurations – saw-tooth aligned and saw-tooth offset. For comparison, the theoretical friction factor values are also plotted. We are interested in seeing the effect of roughness as well as the roughness orientation on f_{cf} .

Figures 24 and 25 show the variation of friction factor for the two cases with air and water flow. It is seen that the data does not agree with the theoretical laminar and turbulent cases and lie significantly above the predicted friction factors. It should be noted that the theoretical friction factors are based on the maximum flow area based on the root height, b .

Figures 26 and 27 show the same data presented in Figs. 24 and 25 for air and water, but the theoretical friction factors are calculated with the constricted flow area based on b_{cf} . It is seen that the agreement between the experimental data and the respective laminar flow equation is significantly improved, (to within less than 5 percent before transition appears to set in). The modified Moody diagram is thus seen to be applicable with the constricted flow area for both saw-tooth and aligned roughness features.

In the turbulent region, the agreement is quite poor. The nature of the roughness elements also makes a difference. The flow does interact with the walls even at the high relative roughness values. Using the average roughness $Ra = 17 \mu m$, the relative

roughness value is calculated as $17/953 = 0.018$. Although this is less than 0.05, the conventional Moody plot is not applicable. It clearly points out the need to introduce new parameters. One of the parameters is the peak height of the element, b . Setting $\varepsilon=b$ is another simple way to represent the roughness elements and is used in Figs. 24 and 25. However, the aligned and offset geometries are not represented by this parameter. Additional parameters need to be looked at as presented in Figs 9-15. A surface profile for the channel is shown in Figs. 2 and 21 along with some of the parameters listed at the top of the chart. A large set of data with different surface features is needed to develop more specific set of parameters suitable for characterization of roughness effects. Similar results are obtained for different channel heights.

CONCLUSIONS

1. The constricted flow area has been introduced as an important parameter in representing the effect of roughness on friction factor. The Moody diagram re-plotted with this parameter results in a constant friction factor in the fully developed turbulent flow region.
2. The effect of surface roughness features is studied with aligned and offset saw-tooth surfaces in a rectangular geometry. The height of the saw-tooth was used to represent the roughness ε . The height of the channel was systematically varied to obtain friction data for air and water for ε/D_h of up to 14%.
3. In the laminar region, the use of the constricted flow area resulted in a significantly improved agreement (within 5 percent, prior to somewhat early

transition to turbulent flow) with the laminar flow data for air as well as water in both aligned and offset configurations.

4. The laminar-to-turbulent transition is not affected appreciably as the channel hydraulic diameter decreases for channels with relative roughnesses below 0.05. However, the transition occurs at lower Reynolds numbers as the relative roughness increases, especially above 0.05.
5. In the turbulent region, the experimental data was considerably above the values obtained using the constricted flow area. The effect of surface features is thus seen to influence the friction factor. Additional data is needed to further quantify this effect.
6. Six parameters are introduced as possible candidates in representing the roughness effect on friction factor in the rough region with $\varepsilon/D > 0$. Further experimental data and analysis is needed before recommending any specific set to represent the flow effects of surface features.

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Table 1. Minichannel Roughness Parameters

	Ra (μm)	Rq (μm)	Rsk	Rku	Rz (μm)	p/e
Smooth	0.21	0.25	0.19	3.07	1.22	-
Aligned	17.0	20.8	1.19	3.45	72.9	6.86
Offset	17.0	20.8	1.19	3.45	72.9	6.86

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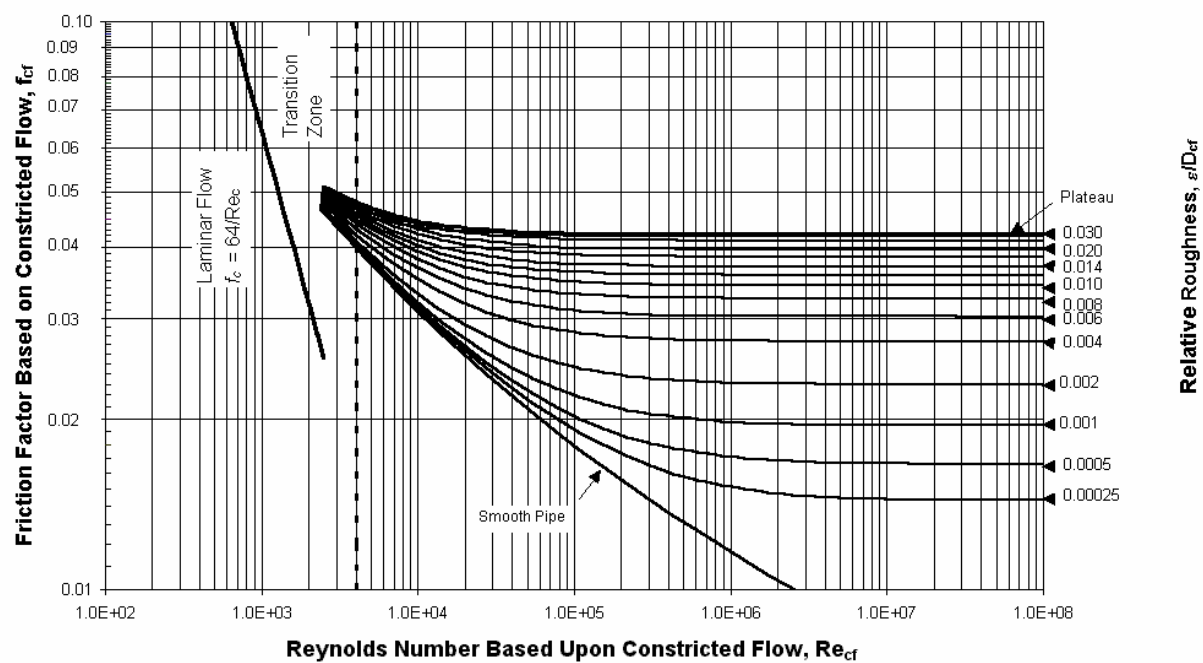


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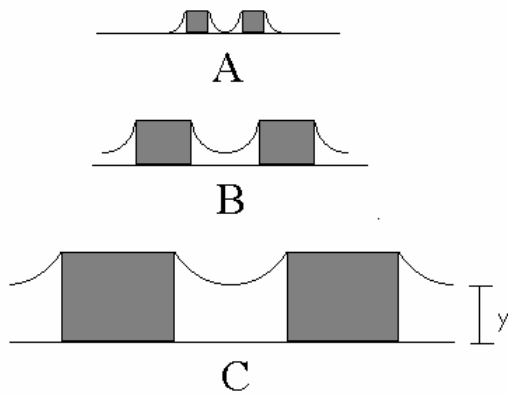


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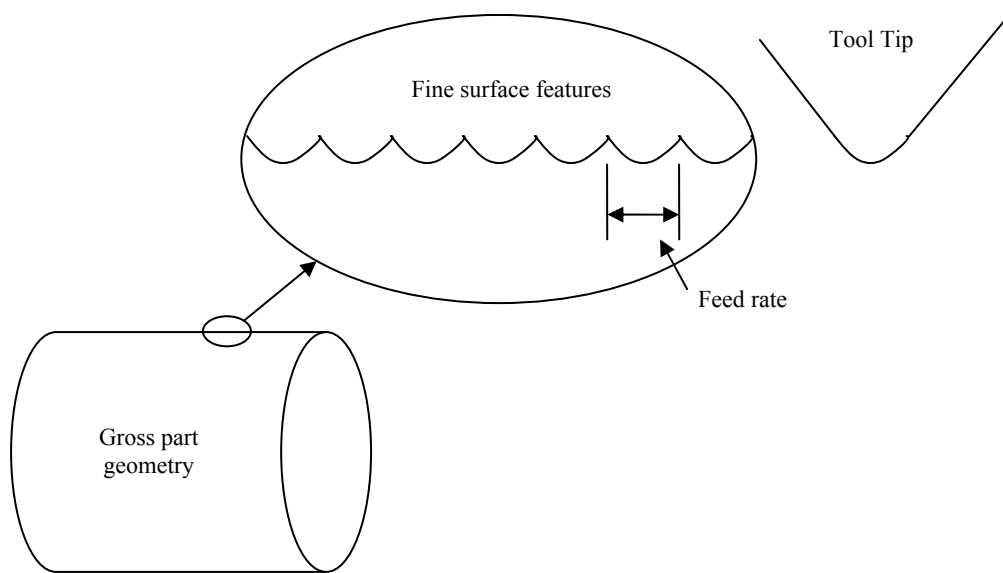


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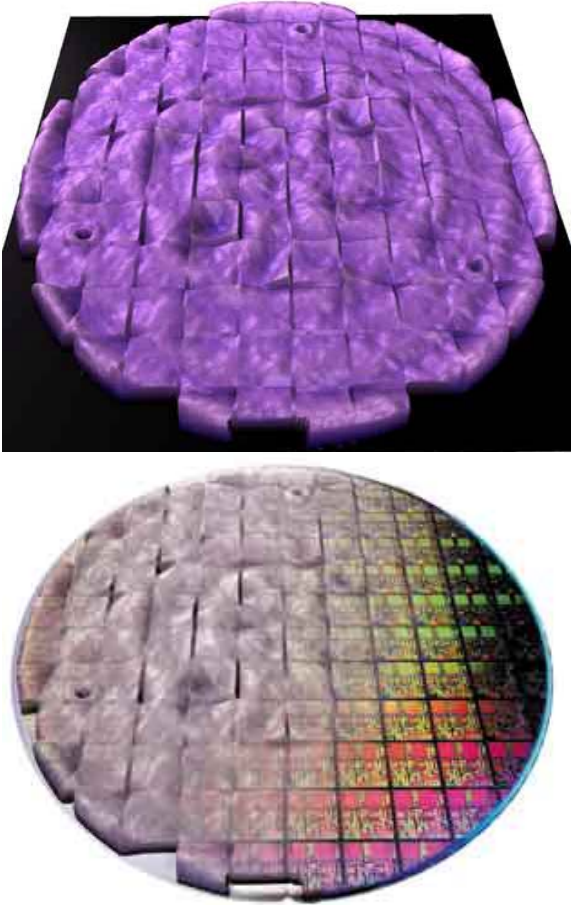


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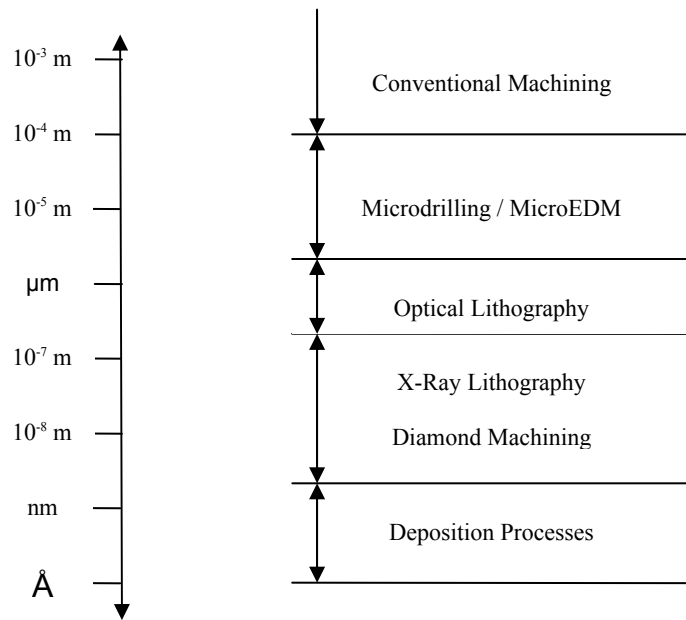


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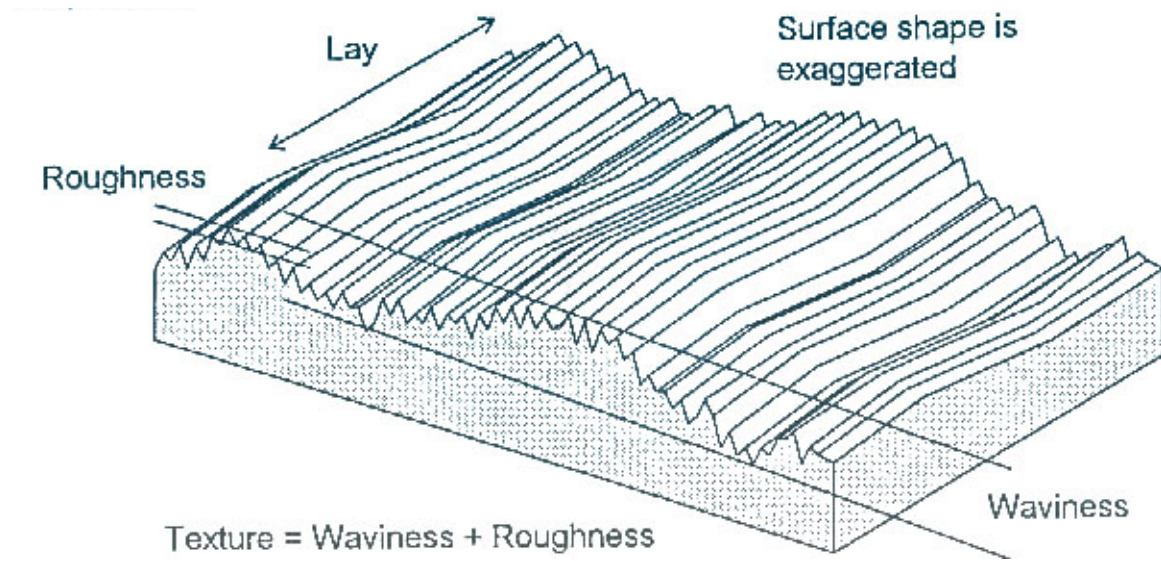


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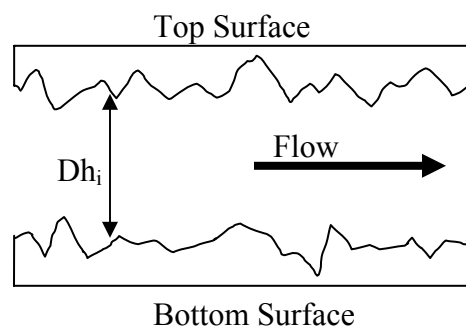


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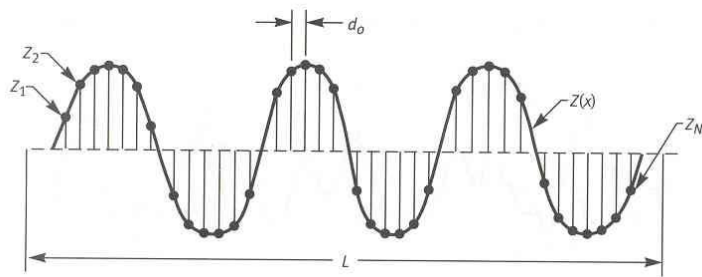


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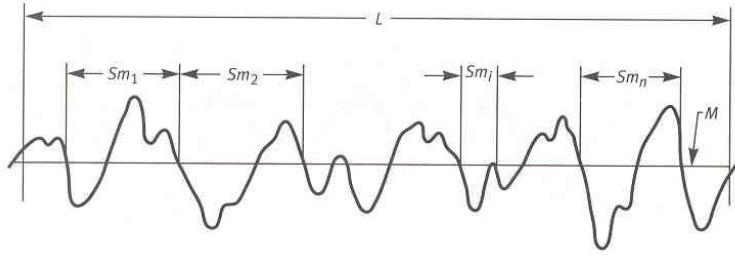


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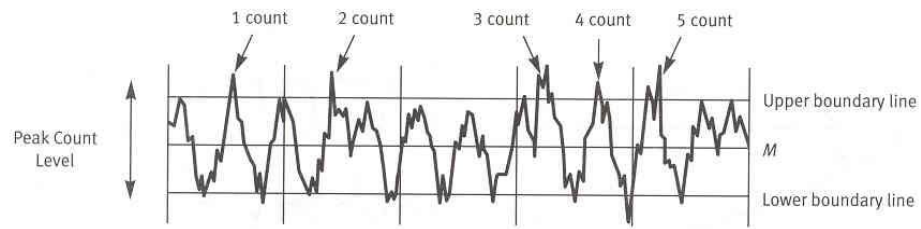


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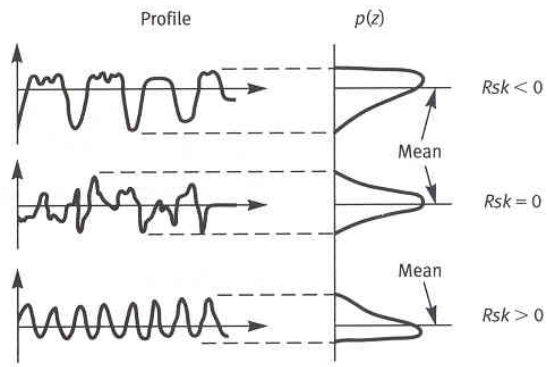


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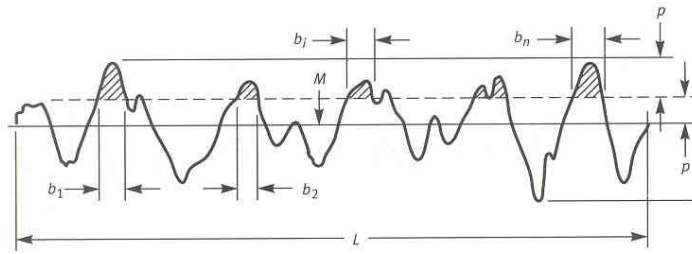


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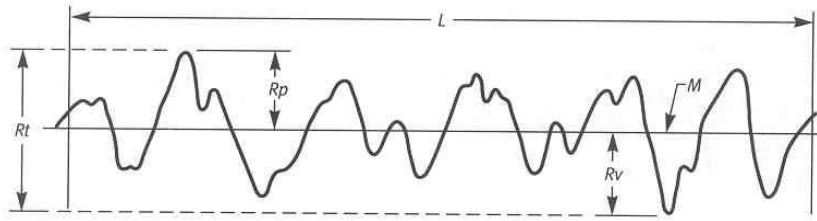


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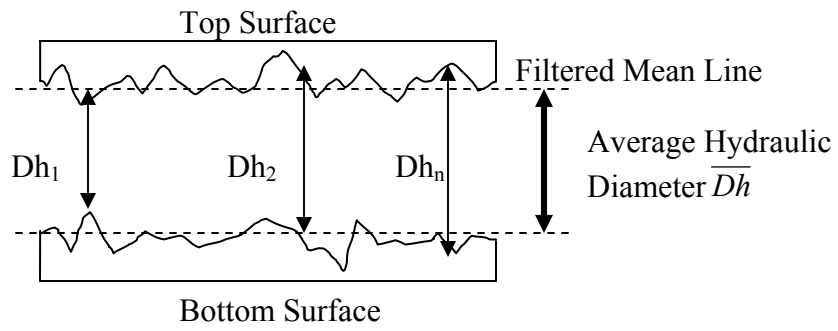


Figure 15: Varying hydraulic diameter along the evaluation length [8]

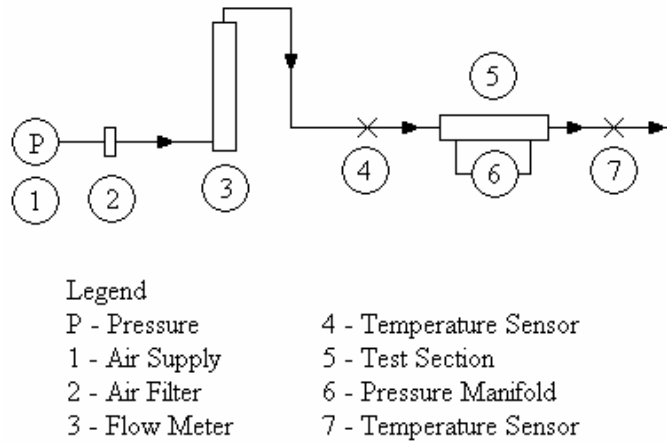


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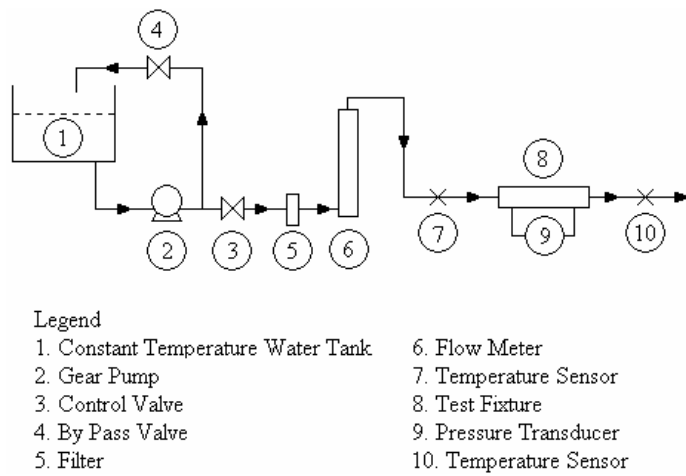


Figure 17: Schematic of Test Setup, Degassed DI Water

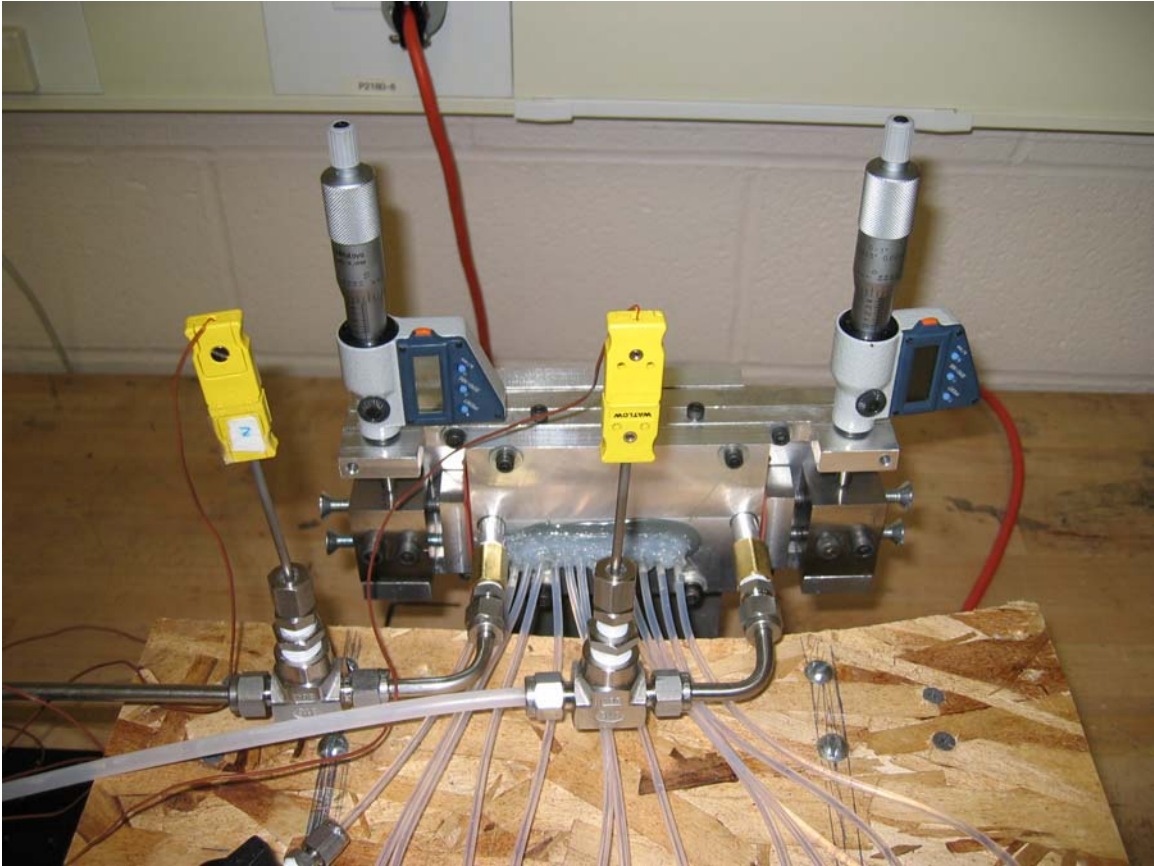


Figure 18 Photograph of the test section assembly

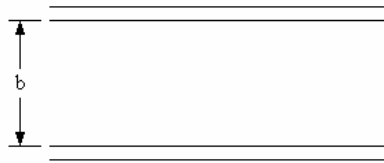


Figure 19: Smooth Channel

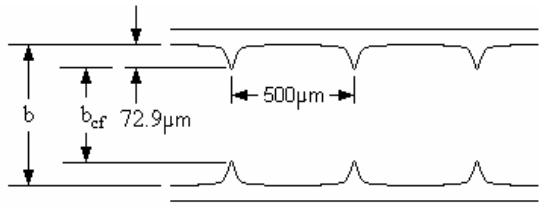


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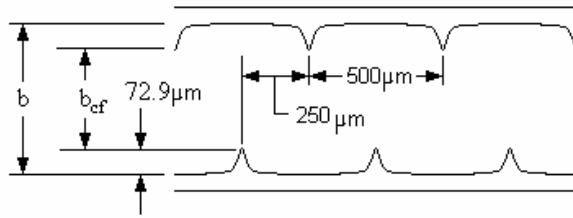


Figure 21: Offset Saw-tooth Roughness

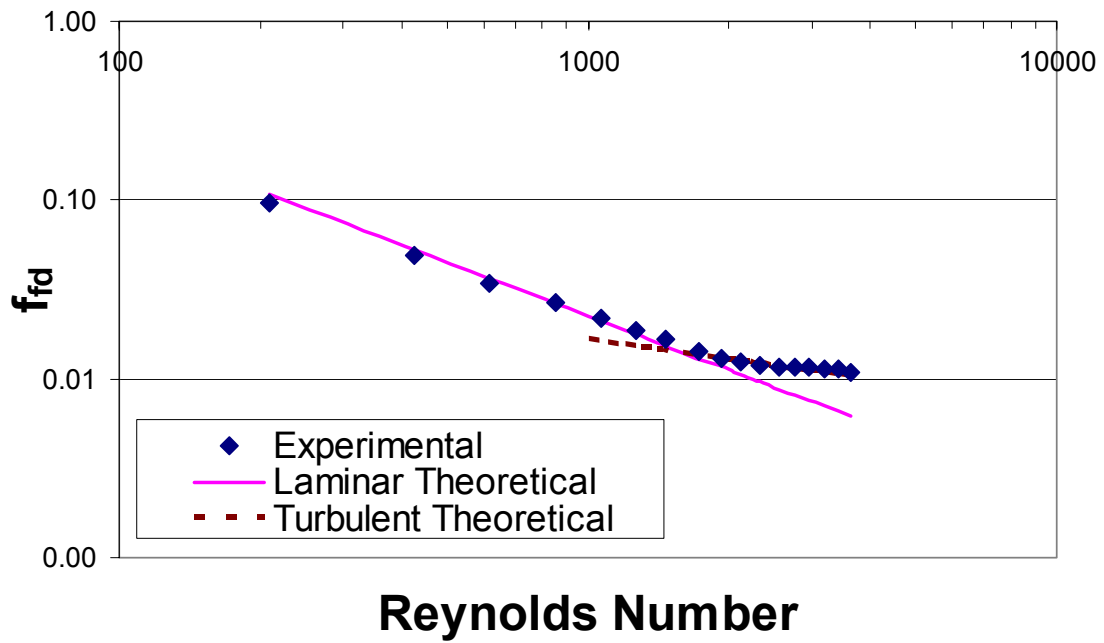


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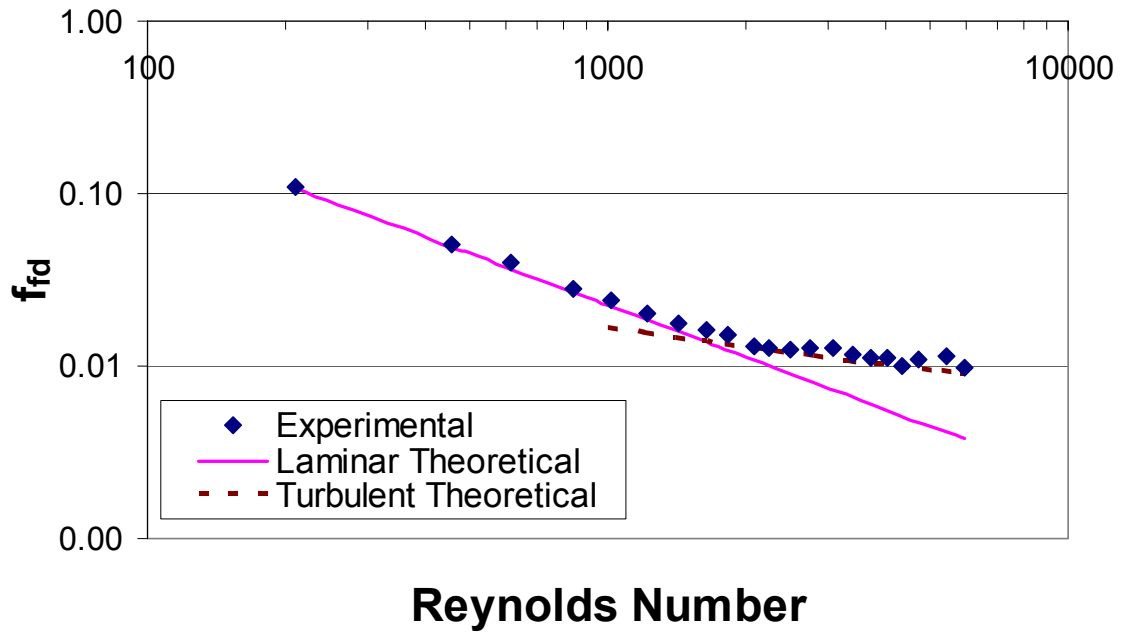


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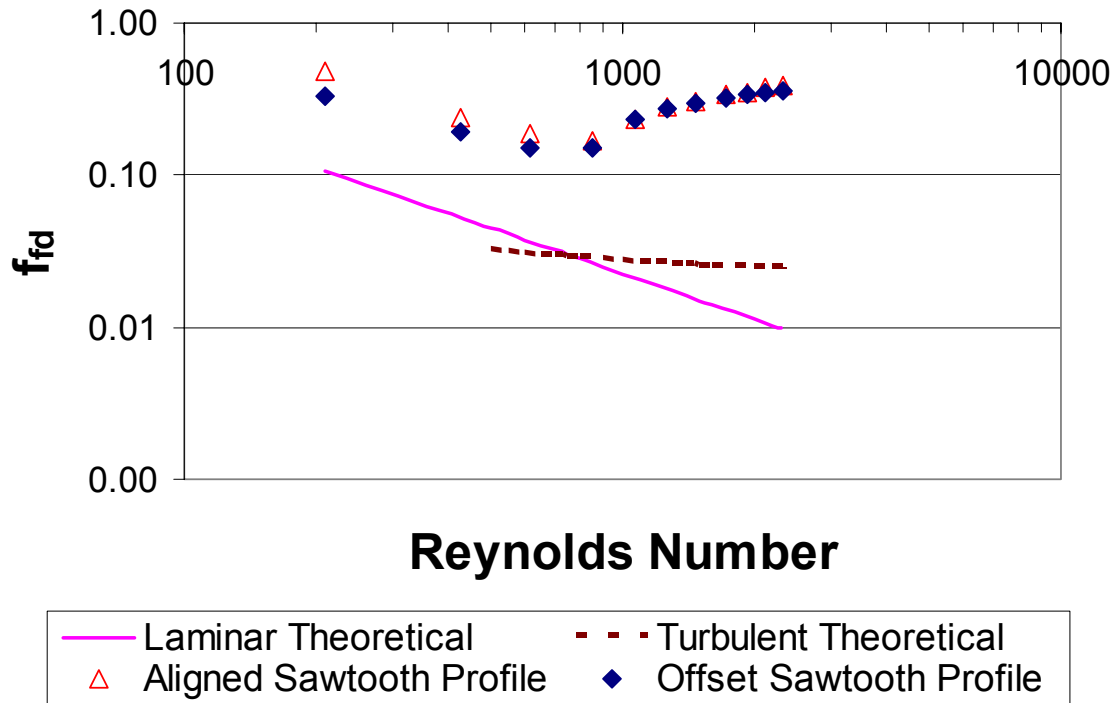


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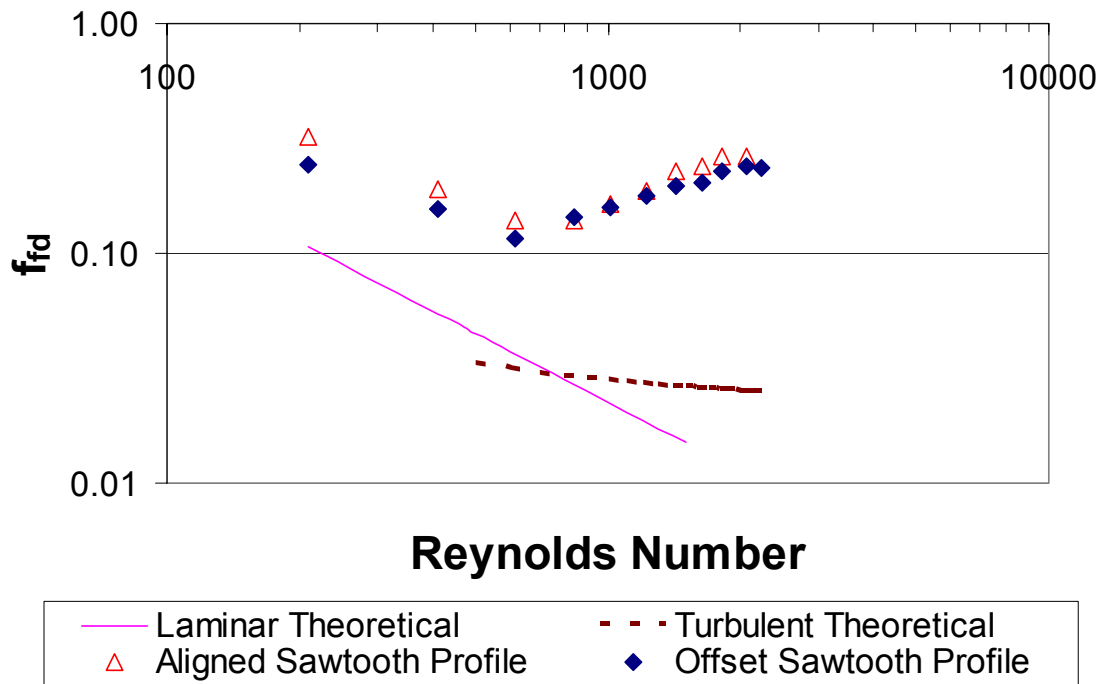


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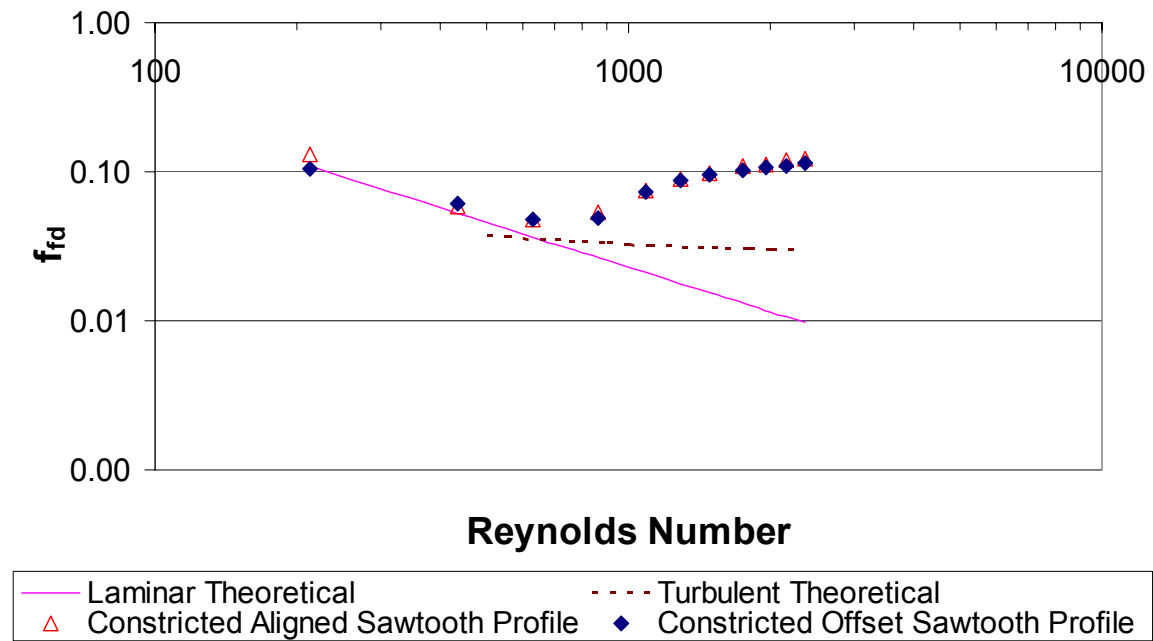


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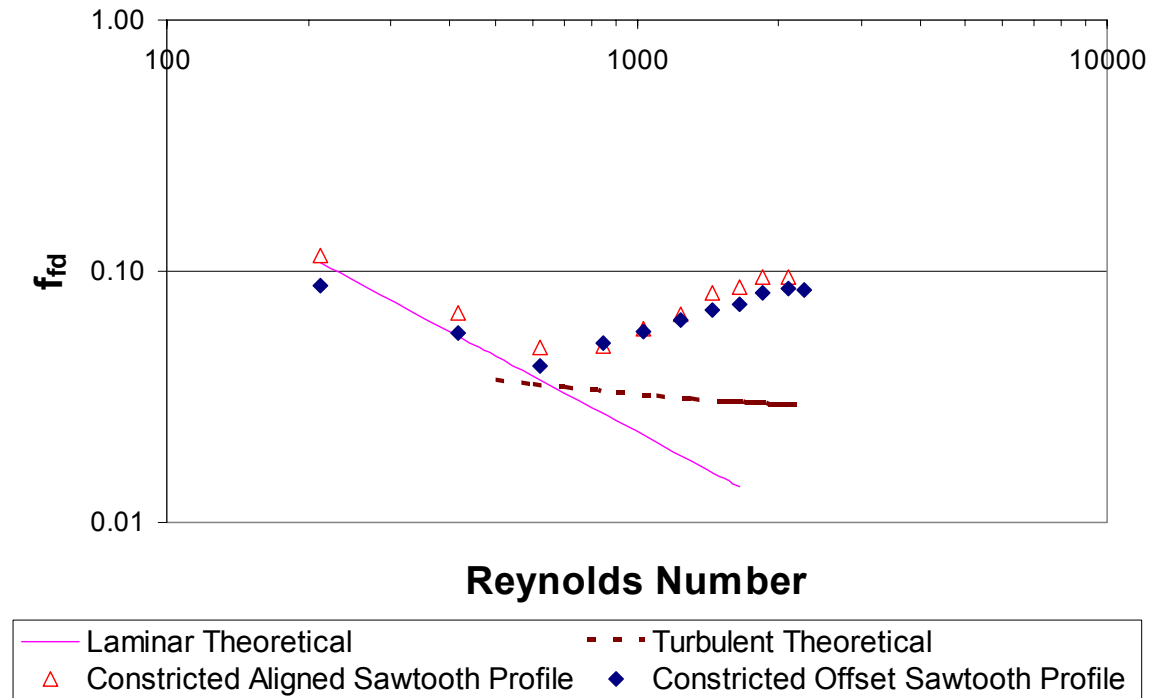


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